

END-TERM PROJECT-02

Game Programming with C++ (CSE 3545)

Programme: B.Tech., CSE

Semester: 6th

Posted on: 09/04/2026

Submission deadline: Last week of Endsem

Subject Learning Outcomes/ Course Outcomes	*Taxonomy Level	Question Number	Marks
Write, compile and execute simple C++ programs to solve computational problems.	L6		10
Become familiar with basic games using SFML library			
Implement Game Mechanics and Physics.			
Apply advanced object-oriented programming concepts like inheritance, polymorphism, and abstraction			
Develop advanced game features like sprite sheets, vertex arrays, texture management, and layering views.			
Refine the game using file I/O, sound effects, HUD & UI.			

*Bloom's taxonomy levels: Knowledge (L1), Comprehension (L2), Application (L3), Analysis (L4), Evaluation (L5), Creation (L6)

Flappy Bird Game

Develop a 2D side-scrolling game using C++ and SFML, inspired by the classic Flappy Bird concept. In this game, the player controls a bird that continuously falls due to gravity. The player must press a key to make the bird “flap” upward and navigate through gaps between vertically aligned obstacles (pipes). The objective is to survive as long as possible and achieve the highest score by passing through obstacles without collision.

Basic Set of Features and Game Mechanics

1. Apply Object-Oriented Programming concepts: Classes like Bird, Pipes
2. Bird Mechanics
 - The bird should continuously fall due to gravity.
 - Pressing a key (e.g., spacebar) makes the bird jump upward.
 - Smooth motion using velocity and acceleration.
3. Obstacle (Pipe) System
 - Pipes appear from the right and move left.
 - Each pipe pair has a vertical gap for the bird to pass through.

- Pipes should be generated at random heights.

4. Collision Detection

- Detect collision between bird and pipes.
- Detect collision with ground or top boundary.
- End game on collision.

5. Score System

- Increase score each time the bird successfully passes a pipe.
- Display score in real-time.

6. Game Window

- Fullscreen or fixed resolution window.

7. Difficulty level

- Easy, Medium, Hard with respect to i) vertical gap sizes between pipes ii) Moving pipes with vertical oscillation

8. Audio & Visuals

- Flap sound effect
- Improved graphics using textures
- Collision sound
- Background music

9. Game Functionalities

- Pause and resume functionality
- Restart option after game over
- High score saved using file handling

Deliverables

1. Source code

2. Executable file

3. Short report including:

- Game design
- Features implemented
- Challenges faced

Sample Output of the Project

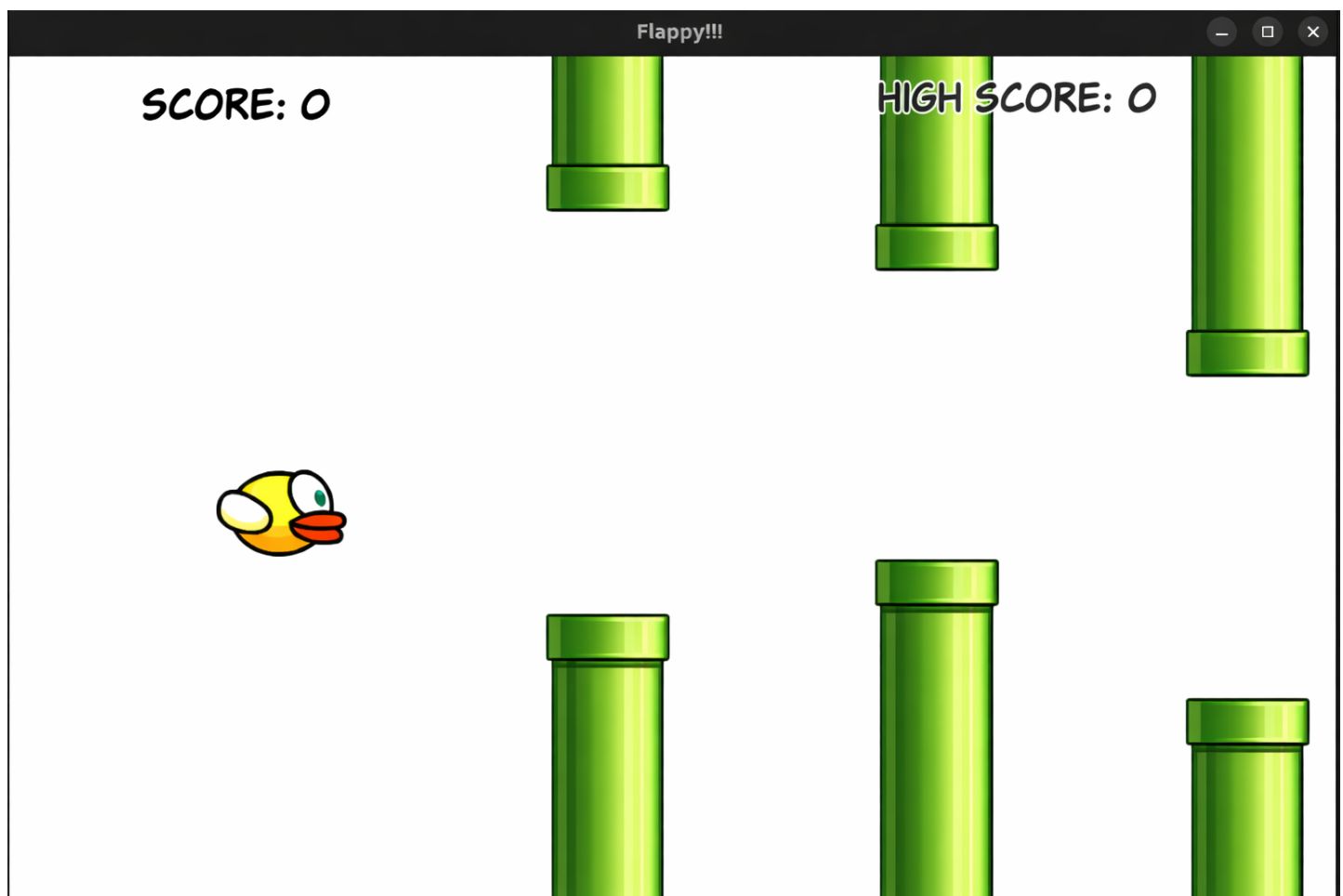


Figure 1: Playable game Screen

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs)

COs	POs											PSOs	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO1	3	2			2						2	3	2
CO2	2		2		3							3	2
CO3	3	3	3	2								3	2
CO4	3	2	3					2			2	3	2
CO5	3		3	2	3							3	3
CO6	2		2		3				2		3	3	3